POWER PULSE

Transmission Line Parameters Important Formulae

Notes by Balaji Sir

1 Resistance and inductance

1.
$$R = \frac{\rho l}{A}, \qquad \frac{R_2}{R_1} = \frac{T + t_2}{T + t_1}$$
temperature effect

2.
$$L_{int} = \frac{\mu_0}{8\pi} = 0.5 \times 10^{-7}\ \text{H/m}, \qquad r' = 0.7788\, r$$
GMR of a solid round conductor

3.
$$L_{1\text{-}\varphi\ \text{loop}} = 4 \times 10^{-7} \ln \frac{D}{r'}\ \text{H/m}$$

4.
$$L_{3\text{-}\varphi,\ \text{per phase}} = 2 \times 10^{-7} \ln \frac{GMD}{GMR}\ \text{H/m}, \quad GMD = \sqrt[3]{D_{12} D_{23} D_{31}}$$
transposed line

5.
$$GMR_{bundle} = \sqrt{r' d}\ (2), \ \sqrt[3]{r' d^2}\ (3), \ 1.09\sqrt[4]{r' d^3}\ (4)$$
bundled conductors

2 Capacitance

6.
$$C_n = \frac{2\pi\varepsilon_0}{\ln(GMD/r)}\ \text{F/m} = \frac{0.02413}{\log_{10}(GMD/r)}\ \mu\text{F/km}$$
use actual radius r (bundle: r_b = √(r d))

7.
$$C_{ab} = \frac{\pi\varepsilon_0}{\ln(D/r)} = \frac{C_n}{2} \qquad (1\text{-}\varphi\ \text{line})$$

8.
$$X_L = 2\pi f L, \quad X_C = \frac{1}{2\pi f C}, \quad I_c = 2\pi f C\, V_{ph}$$
charging current

3 Other points

9.
$$S = \frac{w l^2}{8T}$$
sag (level supports)

10. Skin effect: $R_{ac} > R_{dc}$; ground raises C, lowers L slightly
transposition balances the phases

11.
$$\varepsilon_0 = 8.854 \times 10^{-12}\ \text{F/m}, \quad \mu_0 = 4\pi \times 10^{-7}\ \text{H/m}$$